

# Víctor Manuel Chávez Pérez

## Professor–Researcher · Applied Physics & Science Education

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National Researcher System · Candidate (SECIHTI) 2025–2029



## PROFILE

I am a physicist from the University of Guadalajara and hold a PhD in Applied Physics from the University of Vigo. My research blends atmospheric and climate physics with large-scale data analysis: I study sudden stratospheric warmings (SSW), the Brewer–Dobson circulation and their response under climate-change scenarios using models such as CMIP5 and WACCM.

My work also spans physical oceanography, remote sensing and applied optics. Alongside research, I co-founded the Tlakakini Science & Technology Club, devoted to science outreach for children. I believe the best science is the kind that gets shared.

## RESEARCH INTERESTS

Sudden stratospheric warmings · Stratospheric dynamics · General circulation models (CMIP5 / WACCM) · Climate change · Physical oceanography & remote sensing · Applied optics · High-performance computing · Science education & outreach · Generative AI in science education · Machine Learning applied to climate & atmospheric data · Student perception of AI-based learning tools

## EDUCATION

- 2017 – 2024**    **PhD in Applied Physics**  
Universidade de Vigo · Spain  
Thesis: Sensitivity of sudden stratospheric warmings to simulation and analysis conditions in models.
- 2011 – 2012**    **MSc in Climate Science: Meteorology, Physical Oceanography & Climate Change**  
Universidade de Vigo · Spain  
Temporal and spatial variability of the Minho river plume (MODIS 250 m). Top honors.
- 2009 – 2011**    **MSc in Hydrometeorology — Meteorology & Physical Oceanography**  
Universidad de Guadalajara · Mexico  
Seasonality and anomalies of oceanographic variables in the Gulf of Mexico via remote sensing.
- 2010 – 2011**    **Diploma in Teaching Competencies**  
Universidad de Guadalajara · Mexico  
Pedagogical training for university teaching.
- 2000 – 2008**    **BSc in Physics**  
Universidad de Guadalajara · Mexico  
Thesis: Ocean and coastal meteorology of Nayarit (San Blas, 1999–2002). Grade 95/100.

## EXPERIENCE

- 2025 – present**    **Lecturer «B»**  
Centro Universitario de Tlaquepaque · UdeG · Tlaquepaque, Mexico  
Department of Basic Sciences: Physics and Mathematics.
- 2025 – present**    **Lecturer**  
Centro Universitario Bonpland y Humboldt · Online  
PhD in Aerospace Engineering: Research Methodology and Fluid Mechanics.
- 2019 – present**    **Lecturer «B»**  
Universidad Tecnológica de Jalisco · Guadalajara, Mexico  
Mechatronics Division — Basic Sciences.
- 2016 – 2024**    **External Collaborating Researcher**  
Universidade de Vigo · EPhysLab · Ourense, Spain  
Doctoral research on sudden stratospheric warmings in CMIP5 and WACCM models; statistical analysis of large climate datasets.
- 2012 – 2016**    **Predoctoral Researcher (FPI Fellowship)**  
Universidade de Vigo · EPhysLab · Ourense, Spain  
Statistical analysis of large-volume data for climate-change research with general circulation models.

- 2019 – 2020**    **Faculty Lecturer**  
Tecnológico de Monterrey · Campus Guadalajara · Guadalajara, Mexico  
High school: Matter & Energy, Integral Calculus, Science & Technology.
- 2019**    **Physics Instructor**  
UNITEC · Campus Guadalajara · Guadalajara, Mexico  
Engineering programs: Statics.
- 2012 – 2013**    **Physics & Math Instructor**  
Universidad de Especialidades · Guadalajara, Mexico  
Mechanics, Introduction to Physics and Electromagnetism.
- 2008 – 2011**    **Professor–Researcher**  
Instituto Tecnológico Superior de Zapopan · Zapopan, Mexico  
Mathematics for engineering; parallel computing applied to oceanography.

## PUBLICATIONS

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- 2025**    **Observación Astronómica en México: Pasando de lo Visible a lo Invisible**  
Almaguer-Gómez C., Medina J. F., Chávez-Pérez V. M., et al. · *Óptica Pura y Aplicada* 58(1):15 · DOI: 10.7149/OPA.58.1.51200
- 2025**    **Influence of Satellite and Presatellite Periods on the Validation of Major Sudden Stratospheric Warmings in Historical CMIP5 Simulations**  
Chávez-Pérez V. M., Añel J. A., et al. · *Atmosphere* 16(5) · DOI: 10.3390/atmos16050628
- 2025**    **Temporal Variability of Major Stratospheric Sudden Warmings in CMIP5 Climate Change Scenarios**  
Chávez-Pérez V. M., et al. · *Climate* 13(10) · DOI: 10.3390/cli13100207
- 2023**    **Estacionalidad y Anomalías del Nivel del Mar en el Caribe por Medio de Datos de Altimetría**  
Palacios Hernández E., Chávez Pérez V. M., Sierra Romero L. L. · *Generis Publishing* · ISBN: 979-8-88676-642-4
- 2022**    **Impact of Increased Vertical Resolution in WACCM on the Climatology of Major Sudden Stratospheric Warmings**  
Chávez-Pérez V. M., Añel J. A., et al. · *Atmosphere* 13(4) · DOI: 10.3390/atmos13040546
- 2018**    **The climatology of dust events over the European continent using data of the BSC-DREAM8b model**  
Mandija F., et al. · *Atmospheric Research* 209:144–162 · DOI: 10.1016/j.atmosres.2018.03.006
- In preparation**
- 2026**    Science Beyond the Classroom: Impact Assessment of a Monthly Science Club During National Non-School Days in Mexico  
Almaguer-Gómez C., Chávez-Pérez V. M., et al.
- 2026**    Student Perception of a Generative AI-Based Calculus Learning Assistant: A Mixed-Methods Study  
Chávez-Pérez V. M., et al.
- 2026**    Assessment of the lightning activity response to aerosol optical depth  
Mandija F., Chávez-Pérez V. M.

## ACTIVE RESEARCH PROJECTS

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- 2026 – actual**    **Flood early-warning system for the Guadalajara Metropolitan Area — Principal investigator**  
Centro Universitario de Tlaquepaque de la Universidad de Guadalajara · Funding: No external funding  
Development of an urban flood early-warning system that combines remote sensing (IAM-UdeG weather radar), hydrological runoff modeling over high-resolution LiDAR terrain, and official flood inventories. The system classifies risk by sub-basin in real time and estimates water arrival time, aiming to calibrate thresholds through a retrospective analysis with historical Civil Protection/Fire Department records.
- 2026 -**    **Generative AI for lab practice reports (Google Gems) — Principal investigator**  
Centro Universitario de Tlaquepaque de la Universidad de Guadalajara · Funding: No external funding  
Implementation of Google Gems as a writing and feedback assistant for students to draft and self-assess their lab practice reports in physics courses.
- 2025 – actual**    **AI-generated educational comics for science teaching — Principal investigator**  
Universidad de Guadalajara · Funding: No external funding  
Development and evaluation of educational comics created with generative AI tools as a didactic resource for teaching physics, mathematics and science at basic and upper-secondary levels.

- 2025 – actual** **Pedagogical Calculus Assistant with Google Gems: development and student perception — Principal investigator**  
 Universidad Tecnológica de Jalisco · Funding: Technological University of Jalisco  
 Design and implementation of a Calculus learning assistant based on Google Gems, with quantitative and qualitative study of students' perception, satisfaction and academic performance.
- 2025 – 2028** **Optical elements to enhance visual acuity via wavefront coding — Collaborator**  
 Universidad de Guadalajara · Funding: Universidad de Guadalajara  
 Design and validation of optical elements that improve visual acuity through wavefront coding.
- 2025 – actual** **World Wide Lightning Location Network (WWLLN) — Collaborator · antenna lead**  
 University of Washington · Funding: University of Washington  
 Global lightning location network. In charge of the receiving antenna/node and real-time data delivery.
- 2024 – actual** **Tlakakini Science & Technology Club — Coordinator**  
 CUCEI · Universidad de Guadalajara · Funding: University of Guadalajara · CUCEI  
 Science-outreach club for children, with monthly experiment and workshop sessions.

## TECH DEVELOPMENTS

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### Flood early-warning system · Guadalajara Metro Area

Python · FastAPI · Leaflet · NumPy · pysheds · rasterio · GDAL · Docker

Real-time web platform that anticipates urban flooding in the Guadalajara Metropolitan Area. It fuses the IAM-UdeG weather radar (one scan per minute) with a hydrological runoff model built on 5 m LiDAR and Copernicus DEM terrain, plus the IMEPLAN official flood inventory (363 recurrent sites). It produces a per-zone risk light —calibrated against official water-depth records— and a continuous risk grid that routes rainfall downslope, with concentration time and a "water incoming" ETA. Includes live and historical-replay views. Deployed in production.

## TEACHING

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**Centro Universitario Bonpland y Humboldt** · Online — Fluid Mechanics, Research Methodology

**Centro Universitario de Tlaquepaque · UdeG** · Tlaquepaque, Mexico — Physics & Mathematics, Programming Principles, Data Science

**Universidad Tecnológica de Jalisco** · Guadalajara, Mexico — Physics & Mathematics, Programming Principles, Data Science

**Tecmilenio · Campus Guadalajara** · Guadalajara, Mexico — Matter & Energy, Integral Calculus, Science & Technology

**UNITEC · Campus Guadalajara** · Guadalajara, Mexico — Statics

**Universidad de Especialidades** · Guadalajara, Mexico — Mechanics, Intro to Physics, Electromagnetism

**CETI Colomos** · Guadalajara, Mexico — Dynamics, Differential Calculus

## SCIENCE OUTREACH

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- 2024 –** **Tlakakini Science & Technology Club**  
 Co-founder. Monthly outreach sessions for children on national non-school days.
- 2009 –** **Physics conferences (AGU, EGU, SMF)**  
 Talks at national and international conferences on climate and the stratosphere.
- 2005 –** **Training science communicators**  
 Optics and electromagnetism workshops; National Outreach Meeting.
- 2009 –** **Environment & Sustainability Thematic Network**  
 Member of the CONACYT environment and sustainability network.
- 2011 –** **Environmental Physics Laboratory (EPHysLab)**  
 External collaborating researcher with the University of Vigo group.

## GRANTS & HONORS

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- 2025–2029** **National Researcher System · Candidate**  
 SECIHTI · Mexico
- 2012–2016** **FPI Grant · Predoctoral Researcher**  
 Ministerio de Educación · Spain
- 2011–2012** **Climate Science Master's Scholarship**  
 Campus do Mar · Spain
- 2011–2012** **«Ourense Exterior» mobility grant**  
 Diputación de Ourense · Spain
- 2009–2011** **Hydrometeorology Master's Scholarship**  
 CONACYT · Mexico

**2009**

## **Admission to the State Research System**

COECYTJAL · Jalisco, Mexico

### **Certifications**

**2025**

AI for Educators — Transform your teaching with ChatGPT — Red de Universidades Anáhuac

**2025**

What is Data Science? — IBM

### **TOOLBOX**

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Python · Fortran · MatLab · IDL · NCL · LaTeX · COMSOL · LabVIEW · MPI · Linux · Moodle · Pandas · Scikit-learn · Jupyter